

Subject Description Form

Subject Code	ITC4043T
Subject Title	Fabric Technology II
Credit Value	3 credits
Level	4
Pre-requisite/ <Co-requisite> / (Exclusion)	ITC2004T Textile Studies II
Objectives	The subject aims to deepen and further the contents of Fabric Technology I so that the student will be better equipped with the essential knowledge to develop into a competent textiles technologist.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. evaluate critically the principles of needle selection for weft knitting machines including electronic needle selection; recognize the relationship between knitting parameters and fabric properties ; b. articulate the essential operations in the production of fully fashioned garments; explore the essential operations in the production of fully fashioned garments; c. identify different types of non-woven technologies from their processing features; types of products and applications of such products; d. outline the advancement of weaving technologies from basic motion developments, special types of weaving machines to the system integration. e. communicate efficiently and clearly with textile professionals
Subject Synopsis/ Indicative Syllabus	(I) Knitting: needle selection principles applied in weft knitting; dimensional properties of knitted fabrics; essential elements of fully fashioned knitting; recently developments in knitting; (II) Weaving: Non-woven technologies including spunbonding, spunlacing, needle punching, stitch and adhesive bonding; different non-woven fabrics and their applications. New advancements of weaving machines and special types of weaving machines such as multiphase and circular weaving machines
Teaching/Learning Methodology	Teaching and learning approach will be similar to those adopted for Fabric Technology I that subject material will be disseminated primarily in lectures which are supplemented by seminars and tutorials. Appropriate demonstration of the fabric production machineries and related tools and instruments will be conducted in laboratories.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d	e	
	1. Coursework	50%						
	2. Final Examination	50%						
	Total	100%						
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Assessment of course work may include written assignments, small project reports, presentation and written tests where appropriate depending on the progress of learning and teaching. This will serve as reinforcement of learning as well as feedback of progress. The final examination is able to provide an overview of the performance of the student in learning.								
Student Study Effort Expected	Class contact:							
	• Lecture		22 Hrs.					
	• Tutorial		4 Hrs.					
	• Laboratory		18 Hrs.					
	Other student study effort:							
	• Assignments		59 Hrs.					
	Total student study effort		103 Hrs.					
Reading List and References	<u>Books</u> Spencer, D., (2001). <i>Knitting Technology</i>, 3rd Edition, Woodhead Publishing Ltd. Iyer, C., (1992). <i>Circular Knitting</i>, Meisenbach Bamberg. Mills, R.W., (1965). <i>Fully-fashioned garment manufacture</i>, London Cassell. Brackenbury Terry, (1992). <i>Knitted Clothing Technology</i>, Blackwell Scientific Publications. Adanur, Sabit, (2001). <i>Handbook of Weaving</i>: Technomic Pub. Co. Inc. Talavášek, Oldřich, (1981). <i>Shuttleless Weaving Machines</i>, Elsevier Scientific Pub. Russell, S, (2006). <i>Handbook of nonwovens</i>, Edited by, Woodhead Publishing. R K Dharmadhikary, T F Gilmore, H A Davis and S K Batra, (1995). <i>Thermal bonding of nonwoven fabrics: Textile Progress</i> Vol. 26 No. 2.							